

RestaurantOS

An all-in-one restaurant management platform — built for Nepal.

Project proposal · May 2026

The problem

Nepali restaurants juggle four or five disconnected tools — a paper KOT pad, a separate billing register, hand-written attendance, spreadsheets for payroll, and ad-hoc payment apps (eSewa, Khalti, POS). Orders get lost in the chaos, customers don't know when their food is ready, and owners can't see what their restaurant earned yesterday without leafing through receipts.

The solution

RestaurantOS is a single web platform that runs the whole front-of-house: QR/table ordering, self-serve kiosks, waiter-tablet mode, a kitchen display, a customer pickup board, integrated payments, immutable receipts, and a manager dashboard with live analytics. Built mobile-first so a phone is enough; scales to mounted tablets and TVs.

What it gives each role



Customers

Scan QR → browse menu → order → pay (CASH / POS / eSewa / Khalti / PhonePay / iBank) → live tracking + printable receipt.



Restaurant Managers

Dashboard with revenue charts & top sellers; orders, tables, menu, payments, employees, attendance & payroll — all in one place.



Platform Admins

Multi-restaurant overview: drill into any restaurant's totals, staff count, last order, best-sellers; suspend or reactivate users.

Headline features



QR & table ordering

Each table has a short code (e.g. BH0-A3F2) and a QR — customers scan, order from their phone, pay live.



Self-serve kiosk

Full-screen touch kiosk for quick-serve counters. One-click checkout, payment, receipt.



Kitchen Display

Tablet-friendly card grid. Pending / Preparing tickets with elapsed-time alerts. One-tap status transitions, undo, and cancel.



Pickup Board

TV-mounted board showing READY tickets in giant emerald digits. Customers spot their daily ticket (100–999) at a glance.



Printable receipts

80mm thermal-printer optimised. Immutable receipt table snapshots restaurant + items + payment — gateway-integration ready.



Live analytics

Daily & weekly revenue bar chart, top 5 sellers, custom date range — all on the dashboard, no extra page.

Tech stack

Backend

Java 17

Spring Boot 3.2

Spring Security · JWT

JPA · QueryDSL

Flyway migrations

MySQL 8

AWS S3 (file uploads)

Frontend

Vue 3 · <script setup>

TypeScript

Vite

Pinia (state)

Vue Router

Tailwind CSS

Lucide icons

QRCode + vue-sonner

Architecture at a glance

Customer

	QR scan	→ /table/<code>	→ order & pay
	Kiosk	→ /kiosk/<code>	→ order & pay
	Pickup board	→ /pickup-board/<c>	→ READY tickets only
	Tracking	→ /track/<orderNum>	→ live status + receipt

Manager

/dashboard Orders · Kitchen · Pickup Board · Payments
Tables · Menu · Employees · Attendance · Payroll
Live Revenue + Top-Seller analytics

Admin

/admin All restaurants · Drill-down per restaurant
Users (suspend / reactivate / role change)

Backend

Spring Boot REST ► Controller → Facade → Service → Repository
12 Flyway migrations · multi-tenant by restaurant_code
Immutable receipt table – ready for eSewa/Khalti webhooks

Delivery status

- **DONE** Landing page · Waiter mode w/ images · Live tracking · Receipt printing
- **DONE** Manager analytics · Admin drill-down · Kitchen Display System
- **DONE** Pickup boards (TV + dashboard) · Daily ticket numbers · Payments + receipts
- **NEXT** Email + SMS / Viber receipt delivery · Customer feedback (star rating)

Why now?

Digital wallets (eSewa, Khalti) are mainstream in Nepal but most restaurants still write bills by hand. The receipt table is already structured to accept signed payloads from those gateways — wiring real eSewa/Khalti integration is

a small additional step, not a rebuild. Multi-restaurant from day one means a single deployment can serve a chain or many independents — recurring SaaS revenue with low marginal cost.